Mahdi Bel Haj Said

AI/ML Engineer | Generative AI · LLM Agents & RAG · Computer Vision

Tunis, Tunisia | Open to relocation | mahdi.haj.said@gmail.com | +216 26 957 057

linkedin.com/in/mahdi-bel-haj-said | github.com/Mahdi-Bel-Haj-Said

SUMMARY

AI Engineer with 2+ years of software engineering experience, including 1.5+ years shipping generative-AI products to production: LLM agents, RAG, LoRA fine-tuning, computer vision and FastAPI services on AWS and serverless GPUs. Holder of an engineering degree in Computer Science (MSc equivalent), with high distinction. Available immediately for a full-time role.

EDUCATION

University of Tunis El Manar (UTM) — Tunis, Tunisia

Engineering Degree in Computer Science (MSc equivalent) — High Distinction 09/2023 – 09/2026

National Bachelor's Degree in Computer Engineering — high distinction, ranked 8th/110 09/2020 – 06/2023

Activities: IEEE Computer Society Chapter (3 years); OPTIMA Junior Entreprise (2 years)

PROFESSIONAL EXPERIENCE

AI Developer Intern at Defendr.gg — Tunis, Tunisia (Hybrid) 02/2026 – 08/2026

- Built AI Poster Studio, a generative-AI system that turns a single JSON input into a publish-ready esports poster in about a minute, at $0.10 per poster: GPT-4o Vision writes the image prompt and gpt-image-2 composes the final design.
- Fine-tuned Qwen-Image (20B diffusion model) with LoRA (PyTorch) and served it from a custom RunPod serverless GPU worker, generating the background library every poster is built on.
- Shipped it end to end as a multi-tenant SaaS (FastAPI, RQ/Redis, MongoDB, Cloudflare R2, React): live on Defendr's platform, where first-month revenue covered the full development cost, and sold to a client through a paid API key.
- Built Scout, a production LLM agent (Groq, gpt-oss-20B) that replaces manual organisation and tournament-creation forms with a conversation and answers esports and platform questions grounded in Defendr's FAQ and live data (RAG).

AI Developer (Part-Time) at Gamefy Academy — Remote 12/2025 – 02/2026

- Developed an AI-powered esports highlight generator (Python/FastAPI, Hugging Face Transformers, Whisper, BART and the Twitch API) that automatically extracted key moments from match VODs and assembled highlight reels.
- Built an ML pipeline for audio transcription (speech-to-text), event detection, and semantic search over match transcripts using sentence-transformer embeddings, reducing manual highlight-review time by about 70%.

AI & Web3 Developer at Defendr.gg — Tunis, Tunisia (Hybrid) 08/2025 – 11/2025

- Built a computer-vision module (OpenCV preprocessing, YOLOv8 detection) that identifies the match winner from user-submitted screenshots with 88% accuracy, adapted to each game's result screen and served through a FastAPI service.
- Implemented the conversion of platform-wallet balances into crypto using NFTs (Solidity smart contracts, ethers.js), including gas-fee handling.
- Created a gamification module that rewards users with platform coins based on their activity inside and outside the platform through external game APIs, redeemable as crypto through the NFT conversion.

Python Developer Intern at Carthage Industries — Tunis, Tunisia 07/2025 – 08/2025

- Developed an automated document-processing system using Python, OCR, and NLP to extract, validate, and organise business documents, reducing manual data-entry time by 60%.

AI Developer at SCP — Megrine, Tunisia 12/2024 – 05/2025

- Built a PDF-to-Excel automation tool for daily business documents (invoices, import/export files and others): document-type selection, text extraction (Tesseract OCR), classification of the extracted information and automatic Excel filling, reaching 99.7% accuracy and processing 100 documents in 20 minutes, with an optional human-validation step.
- Deployed the full 3-tier system (FastAPI backend, React frontend, PostgreSQL database) on AWS (VPC, EC2, S3, IAM).
- Integrated an employee chatbot (OpenAI API + ChromaDB RAG) answering from the company's database and documents (e.g., exact export figures for a given month), with guardrails restricting it to work-related questions.

AI Development and Embedded Systems Intern at ACTIA — Tunis, Tunisia 06/2024 – 07/2024

- Developed a Random Forest anomaly-detection model (Python, scikit-learn) that analyses system logs and flags abnormal patterns with 71% precision.
- Performed hardware diagnostics and electronic-board testing in the after-sales department; contributed to embedded-systems workshops.

Full Stack Development Intern at HOTIX — Tunis, Tunisia 02/2023 – 06/2023

- Developed an Android mobile application in Java for personalised travel planning, implementing the A* algorithm to generate optimised itineraries from budget, time, and destination constraints.

SKILLS

Programming Languages: Python, JavaScript, TypeScript, C++, Solidity, Java, SQL

Generative AI: LLMs, LLM agents, RAG, LoRA fine-tuning, diffusion models, multimodal LLMs (GPT-4o Vision), prompt engineering, LangChain, OpenAI API, Groq

Machine Learning: Deep learning, NLP, computer vision, OCR (Tesseract), object detection (YOLOv8), speech-to-text, PyTorch, Hugging Face Transformers, sentence-transformers, Whisper, scikit-learn, OpenCV, MediaPipe, Model evaluation, LLM evaluation

Back End & Web: FastAPI, REST APIs, Node.js, Next.js, React, RQ/Redis, Pydantic

Databases & Storage: MongoDB, Redis, Vector databases (ChromaDB), Cloudflare R2, PostgreSQL, MySQL, SQL Server

Cloud & DevOps: Git, Linux, GitHub Actions, Jenkins, RunPod, AWS (EC2, S3, VPC, IAM, Lambda), Docker, SonarQube, Grafana

Spoken Languages: Arabic (native), English (C1), French (C1), Spanish (A2)